

OPERATING MANUAL & MAINTENANCE MANUAL

FJV-200

FJV-250

18000 min⁻¹ SPINDLE

Manual No.: H529SC0560E

Serial No.:

Before using this machine and equipment, fully understand the contents of this manual to ensure proper operation. Should any questions arise, please ask the nearest Technical Center or Technology Center.

IMPORTANT NOTICE

1. Be sure to observe the safety precautions described in this manual and the contents of the safety plates on the machine and equipment. Failure may cause serious personal injury or material damage. Please replace any missing safety plates as soon as possible.
2. No modifications are to be performed that will affect operation safety.
3. For the purpose of explaining the operation of the machine and equipment, some illustrations may not include safety features such as covers, doors, etc. Before operation, make sure all such items are in place.
4. This manual was considered complete and accurate at the time of publication, however, due to our desire to constantly improve the quality and specification of all our products, it is subject to change or modification. If you have any questions, please contact the nearest Technical Center or Technology Center.
5. Always keep this manual near the machinery for immediate use.
6. If a new manual is required, please order from the nearest Technical Center or Technology Center with the manual No. or the machine name, serial No. and manual name.

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1 SAFETY PRECAUTIONS

This manual is prepared only for the function or the device which is specified in the title of this manual. Therefore, you are required to carefully read and understand the operating manual of the machine, especially the section describing safety precautions, before using or operating the function or the device.

1-1 Rules

The meanings of our safety precautions to DANGER, WARNING, and CAUTION are as follows:



DANGER

: Failure to follow these instructions could result in loss of life.



WARNING

: Failure to observe these instructions could result in serious harm to a human life or body.



CAUTION

: Failure to observe these instructions could result in minor injuries or serious machine damage.

1-2 Safety Precautions Relating to the Function/Device Described in This Manual



WARNING

- Never use a tool or insert that exceeds the specification. Failure will cause a serious accident. [Max. insert mass: 20 g (0.04 lbs)]
The operator's door has the strength which can assure safety to the energy effused by a centrifugal force or other external force when an insert mounted around the periphery of a large diameter tool comes off by some reason during a high-speed rotation of the spindle.
- Keep the operator's door closed while the spindle is rotating.
- Shut off the power before changing the oil or applying grease.
Otherwise, you could sustain serious injury.



CAUTION

- When performing a warm up operation, set a balanced high-speed operation tool in the spindle. Otherwise, spindle parts could break (breakage of the bearing, the collet and the rotary joint for through-spindle coolant due to excessive vibration).
- Use only the oil recommended by Yamazaki Mazak.
Otherwise, machine trouble and other problems will occur.

1-3 Precautions

1. Tools rotating at high speed must be carefully balanced. When a tool is used at the speed range of 7000 - 15000 min⁻¹, it must be balanced to the degree the residual imbalance is 10 g·mm (0.07 ft·lbs) or less (recommendation).

When the tool is used at the speed range of 15000 - 18000 min⁻¹, the tool must be balanced to class G2.5 (grade of balancing) or better.

Balancing conditions differ depending on operating conditions such as the spindle speed and the mass of tool.

For details, refer to 1-4 "Balance".

Also observe the items below.

- a. After replacing inserts of a cutting tool, maintain the specified dynamic balance of the tool.
 - b. Never rotate the spindle at a high speed with a poorly balanced tool in it. When rotating the spindle at a high speed, always mount a tool balanced meeting the spindle speed.
 - c. For balancing of high-speed operation tools, consult the manufacturer of the tools.
 - d. Select the tool and the tool holder that meet the high-speed rotation specification referring to the catalog of the tool maker.
2. Every part of the tool is severely affected by the centrifugal force caused by high speed rotation. Since high speed rotation could loosen the screws as well as exert big centrifugal force on clamped object if screws of the tool are not secured tight enough, it is very dangerous.

Note : All screws of the tool must be secured tightly for high speed rotation.

1-4 Balance

Calculations of tool balance

$$G = \frac{\varepsilon n}{9550} \quad \dots(1)$$

$$\varepsilon = \frac{G \times 9550}{n} = \frac{u}{m} \quad \dots(2)$$

$$gr = \frac{u}{r} \quad \dots(3)$$

G : Balance $\left(\frac{\mu\text{m} \times \text{min}^{-1}}{9550} \right)$
 ε : Balancing plane eccentricity $\left(\frac{\text{g} \cdot \text{mm}}{\text{kg}} \right)$
 n : Revolving speed (min^{-1})
 u : Residual imbalance ($\text{g} \cdot \text{mm}$)
 m : Mass of rotor [Mass of tool] (kg)
 r : Correction radius (mm)
 gr : Amount of correction (g)

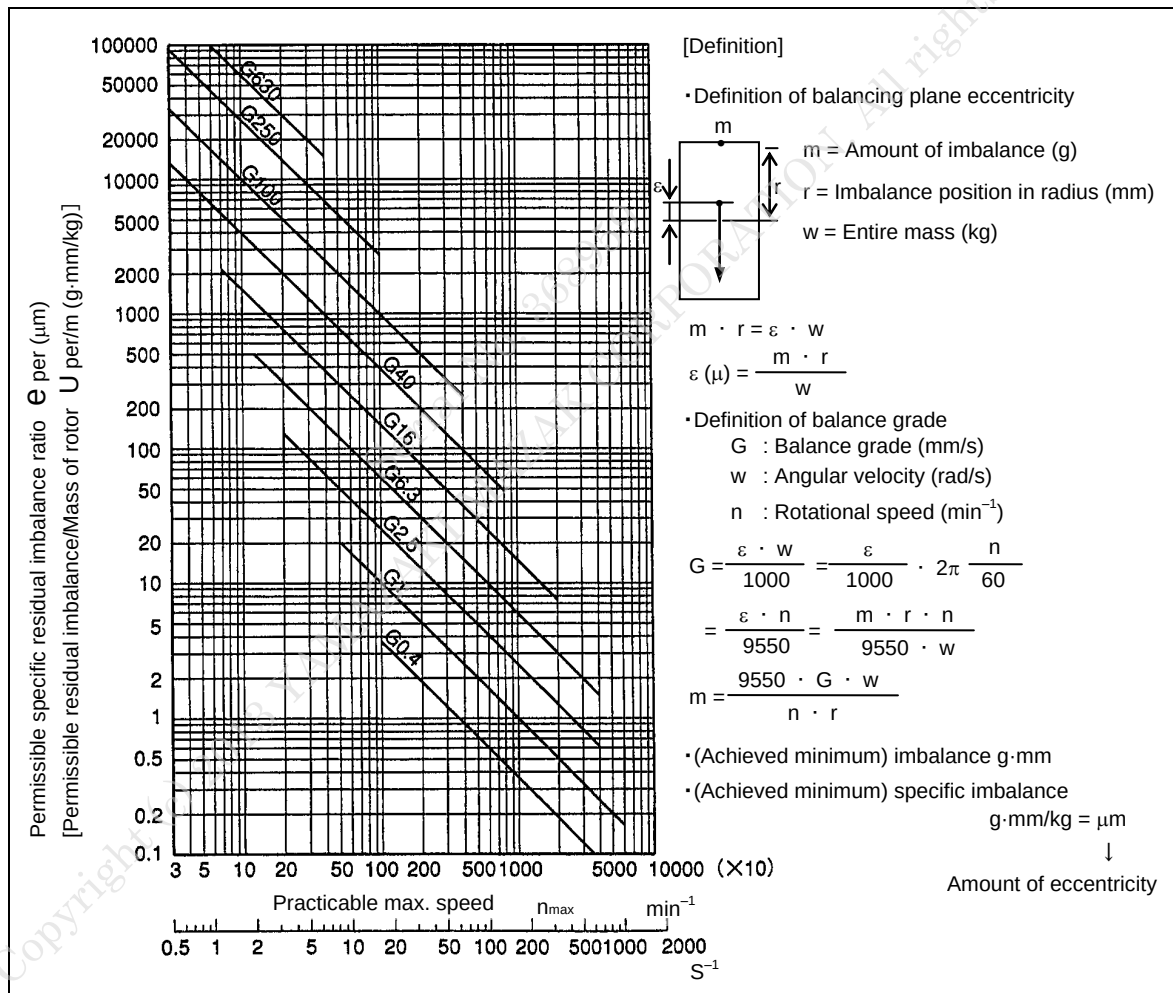


Fig. 1-1 Balance grade and allowable residual imbalance

[Example of calculation]

- Calculation conditions
- Grade of balancing G2.5 (machine tool spindles)
 - Mass of tool: 2 kg (4.4 lbs)
 - Operating speed: 18000 min⁻¹
 - Imbalance correcting radius: 25 mm (0.98 in)

From the graph or formula (2)

$$\text{Balancing plane eccentricity: } \varepsilon = \frac{G \times 9550}{n} = \frac{2.5 \times 9550}{18000 \text{ min}^{-1}} = 1.326 \frac{\text{g} \cdot \text{mm}}{\text{kg}}$$

$$\text{Residual imbalance: } u = m \times \varepsilon = 2 \text{ kg} \times 1.326 \frac{\text{g} \cdot \text{mm}}{\text{kg}} = 2.652 \text{ g} \cdot \text{mm}$$

From formula (3)

$$\text{Amount of correction: } gr = \frac{u}{r} = \frac{2.652 \text{ g} \cdot \text{mm}}{25 \text{ mm}} = 0.11 \text{ g}$$

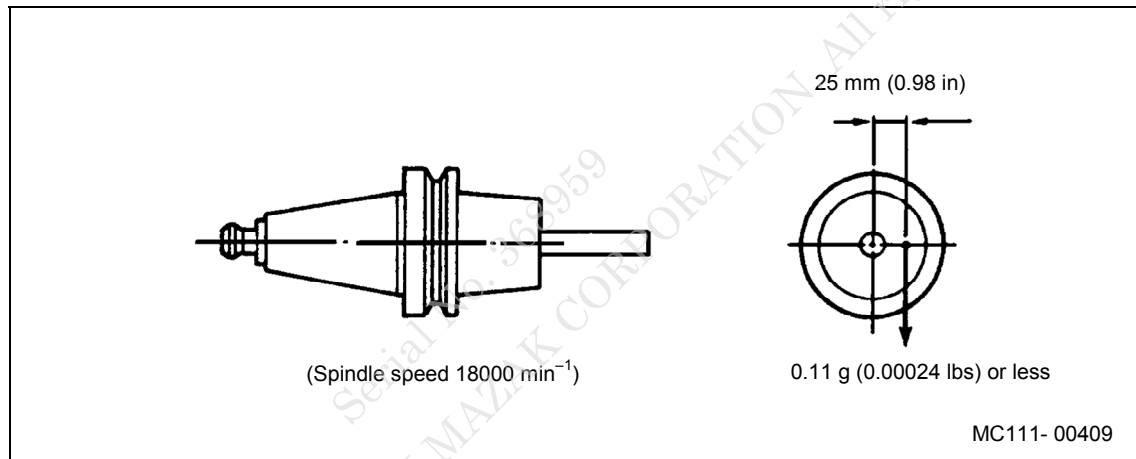


Fig. 1-2 Calculation of dynamic balance (example)

To balance a tool at the user, use the tool balance indicated below.

Recommended tool balancer

Maker	HOFMANN (Germany)*	-
Type	PTB7.1	-
Specification	Imbalance quality	G1.6 or better
	Repeated accuracy	1.0 g·mm or less
	Imbalance correction rate	95% or better
	Display accuracy	0.001 g
	Rotational speed for measurement	400 to 1200 min ⁻¹ (automatic adaptable rotational speed)
	Max. tool mass	7 kg (15.4 lbs)
	Max. tool height	275 mm (10.83 in)
	Max. tool diameter	200 mm (7.87 in)
	Machine dimensions	500 × 460 × 510 mm (19.69 × 18.11 × 20.08 in)
	Voltage	230 V/2.0 A, 115 V/4.0 A
	Software	Balance correction method: Removal amount correction, division correction (3 to 36 parts), etc.
		Automatic correction of adapter mounting error, etc.
		Tooling dimension memory (500 kinds of tooling)
	Protection grade	Safety grade C6 (DIN 45690)

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SAFETY PRECAUTIONS



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2 SPECIFICATIONS

2-1 Major Machine Specifications

Table 2-1 Specification table

Item			Unit	Specifications
Spindle	Spindle tapered bore		—	7/24 taper No. 40 (1/10 taper HSK-A63)
	Tool clamp method		—	Disc spring
	Tool unclamp method		—	Hydraulic piston
	Tool clamping force (Note)		kN (lbs)	11.0 (2425)
	Spindle nose shape		—	ISO BT40 (HSK-A63 / Big-Plus #40)
	Spindle construction		—	Cartridge type
	Spindle bearing	O.D.	mm (in)	100 (3.94)
		I.D.	mm (in)	70 (2.76)
		Width	mm (in)	16 (0.63)
	Spindle speed		min ⁻¹	35 to 18000
	Spindle torque		N·m (ft·lbs)	Max. 120 (88.51)
	Spindle speed command		—	Direct command (1 min ⁻¹ increment)
	Number of spindle speed change ranges		—	2 steps (electric)
	Spindle drive motor	Type	—	AC inverter motor
		Output	kW	30 (50%ED) 22 (Cont. rating)
Machine size	Spindle orientation		—	Electric orientation method (encoder type)
	Spindle tapered hole cleaning		—	Air blast Air source: 0.5 MPa (70 psi) or higher Air source should be prepared by the customer.
	Dimensions (without chip conveyor)	Height (from floor level)	mm (in)	3035 (119.49)
		Width (30-tool magazine)	mm (in)	3010 (118.5)
		Depth	mm (in)	2450 (96.46)
	Mass (standard specification)		kg (lbs)	8500 (18739)
	Power requirement (Standard specification)	50%ED	kVA	50.7
		Cont. rating		62.5

Note : Tool clamping force indicates the force necessary for removing a tool from the spindle. The clamping force is the value set before shipping. The clamping force may be lowered during the use of the machine for a long time.

2-2 Spindle Power/Torque Diagram

1. 18000 min⁻¹, spindle power, torque diagram

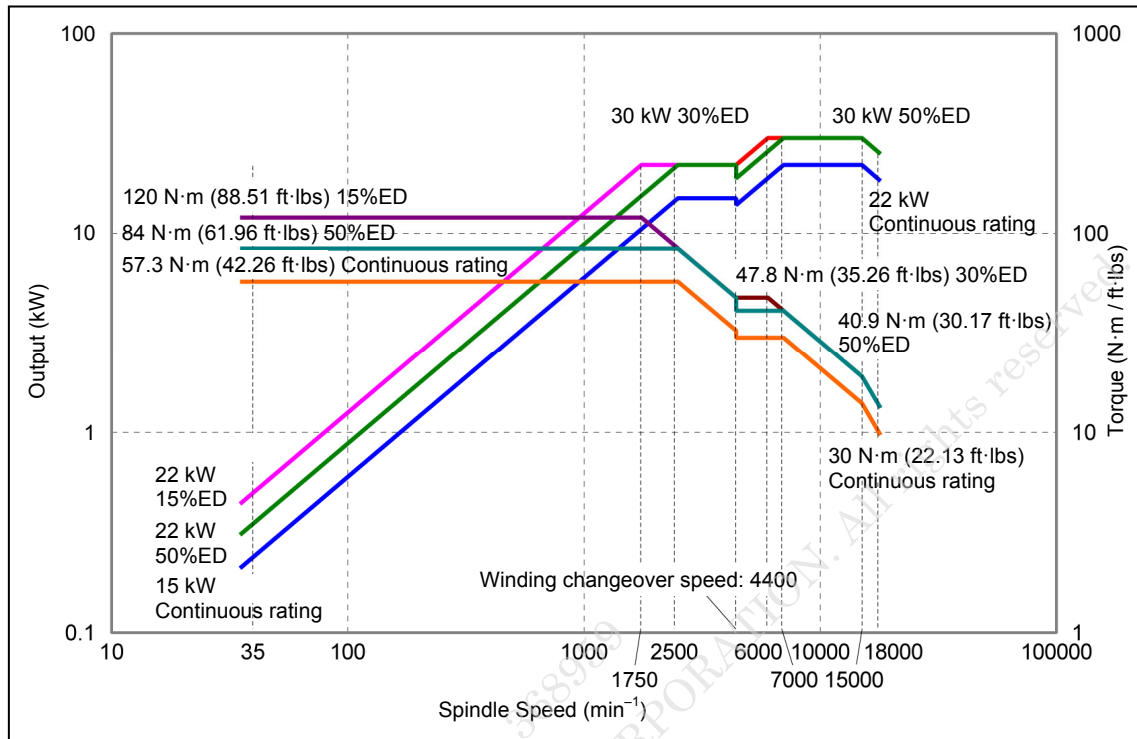


Fig. 2-1 Spindle power/torque diagram 18000 min⁻¹

2-3 Setting the Spindle Speed

The safety spindle speed is set meeting the tool diameter, set as the tool data, to ensure operator's safety against unforeseeable accident such as flying out of an insert from the tool.

For 18000 min⁻¹ specification:

Maximum spindle speed that ensures safety
with the tool of the maximum size set in the spindle: 14349 min⁻¹
Maximum tool diameter that ensures safety with the spindle
rotating at the allowable maximum speed of the machine: $\phi 99$ mm (3.9 in)

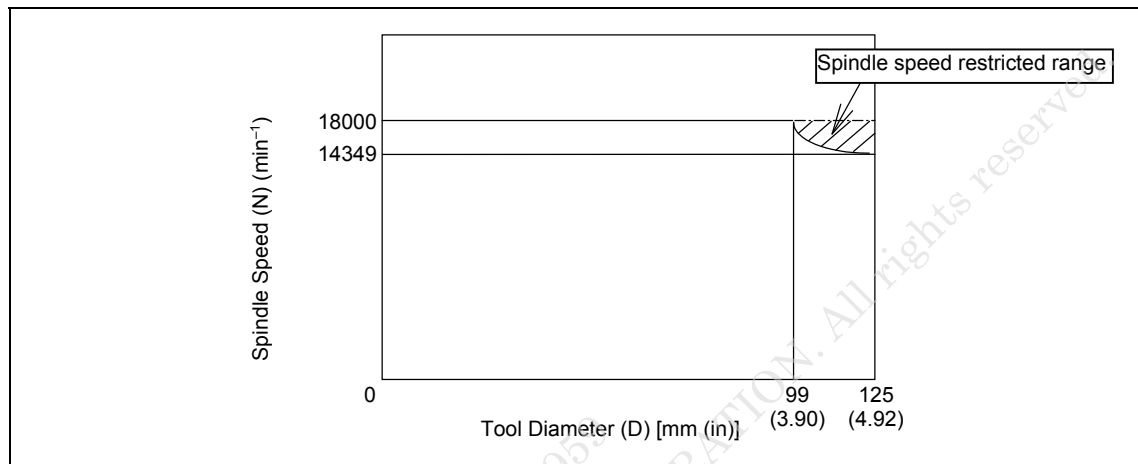


Fig. 2-2 Setting the maximum spindle speed by tool diameter

Note 1: The correct spindle speed cannot be obtained unless the correct tool diameter is input to the tool data.

Note 2: When a value is set for the rotating speed in the tool data, an alarm will be generated if a spindle speed exceeding the permissible speed set for the tool is specified.



- For details, refer to the PARAMETER LIST, ALARM LIST, M-CODE LIST provided separately.



WARNING

- Never use a tool or insert that exceeds the specification. Failure will cause a serious accident. [Max. insert mass: 20 g (0.04 lbs)]
The operator's door has the strength which can assure safety to the energy effused by a centrifugal force or other external force when an insert mounted around the periphery of a large diameter tool comes off by some reason during a high-speed rotation of the spindle.

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SPECIFICATIONS



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3 OPERATION

3-1 Warm Up

Warm up means the spindle run-in operation to be conducted by the user before starting daily operation.

In addition to warm up to be conducted before starting daily operation, it must be conducted in the following cases.

1. After turning on power supply and before starting production (everyday at the start up of operation)

Table 3-1 Warm up conditions

Warm up period	Approx. 30 minutes
Spindle speed	1000 min ⁻¹ → 4000 min ⁻¹ → 8000 min ⁻¹ → 12000 min ⁻¹ → 16000 min ⁻¹ → 18000 min ⁻¹ (for 5 min. at each speed)

2. When power supply is turned off once and then it is turned on again more than 15 minutes after the turning off of power supply
3. When starting the spindle after stopping it for more than 3 hours

Table 3-2 Warm up conditions

Warm up period	Approx. 10 minutes
Spindle speed	6000 min ⁻¹ (2 min.) → 12000 min ⁻¹ (2 min.) → 16000 min ⁻¹ (2 min.) → 18000 min ⁻¹ (4 min.)

3-1-1 Check points during warm up operation

1. Before starting warm up

- Check if the air is flowing out at the spindle nose and on the tapered surface through the discharge holes when a tool is removed from the spindle. If air is not flowing out, clean the discharge ports.
- Check the spindle nose and the tapered surface for adhesion of chips and corrosion. To avoid chip adhesion or corrosion, check the spindle every week.
- The spindle does not rotate unless a tool is mounted in it. Mount a tool in the spindle before performing warm up operation.
- Keep the operator's door closed while the spindle is rotating.



WARNING



CAUTION

- When performing a warm up operation, set a balanced high-speed operation tool in the spindle. Otherwise, spindle parts could break (breakage of the bearing, the collet and the rotary joint for through-spindle coolant due to excessive vibration).

2. During spindle rotation

- Check for the generation of abnormal noise.
- Check for the generation of abnormal vibration.



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4 MAINTENANCE OF MAJOR UNITS

4-1 Construction of the Spindlehead

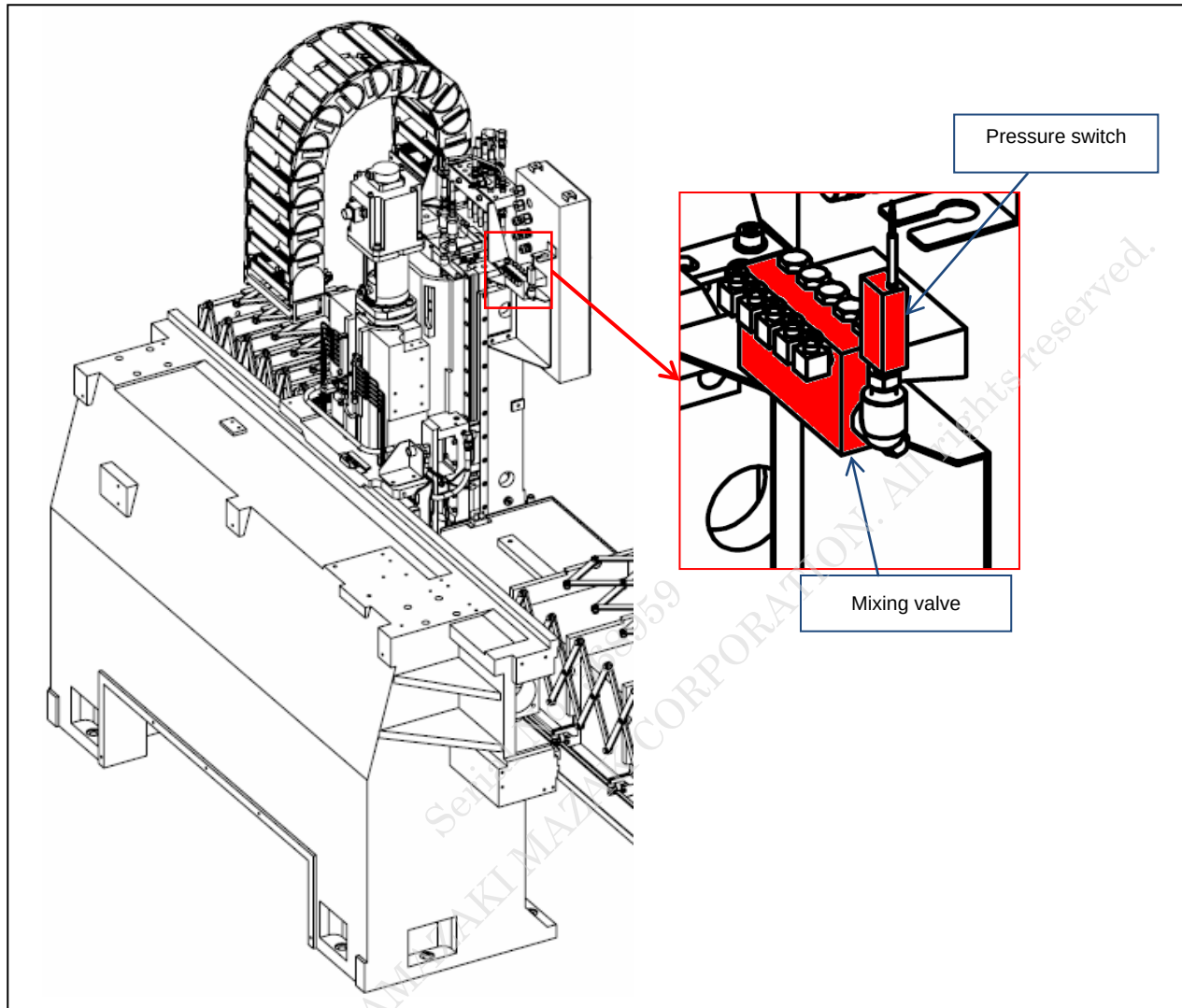


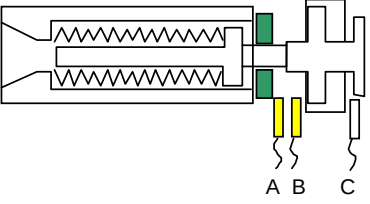
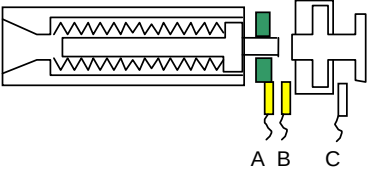
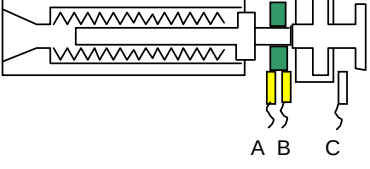
Fig. 4-1 Construction of the spindlehead

- A tool is clamped in the spindle by spring force.
Tool unclamp operation is performed by the hydraulic cylinder.
- The spindle bearings are lubricated by oil-air. The pressure switch is provided to prevent lubrication failure problem even if air pressure is lowered.
- The 18000 min⁻¹ specification spindle uses the sensors that detect if a tool exists or not after the execution of tool clamp operation of the spindle, in addition to the tool unclamp confirmation sensor.

The interlock is taken to disable spindle rotation if there is no tool in the spindle although the spindle executed tool clamp operation.

The relationships between the tool clamped states and the ON/OFF status of the sensors are shown in Table 4-1.

Table 4-1 Tool clamped states and sensor ON/OFF statuses

Tool Clamped State	Diagram	Sensor (A) Clamp with a Tool	Sensor (B) Clamp without a Tool	Sensor (C) Unclamp	Spindle Start
Tool unclamp		OFF	OFF	ON	Spindle start disabled (Alarm)
Clamp with a tool		ON	OFF	OFF	Spindle start enabled
Clamp without a tool		ON	ON	OFF	Spindle start disabled (Alarm)

4-2 Spindle Cooling Circuit

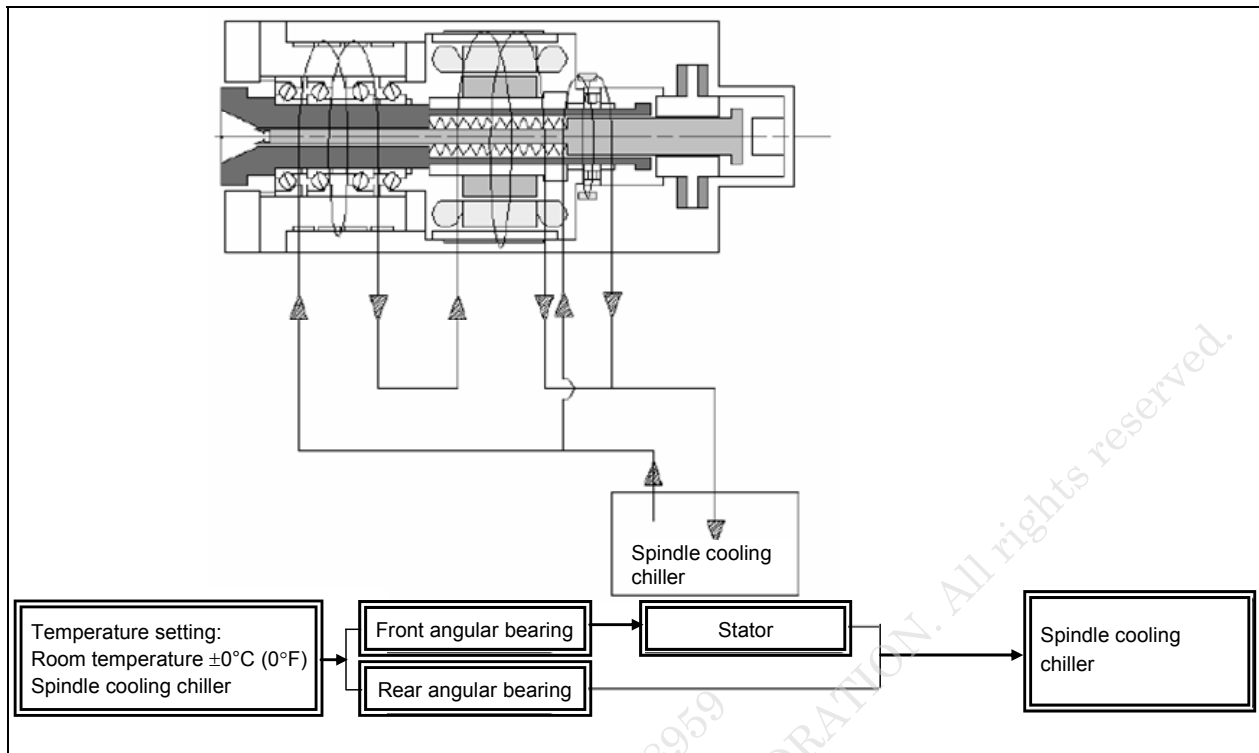


Fig. 4-2 Cooling unit circuit diagram

Coolant temperature is set at $\pm 0^{\circ}\text{C}$ (0°F) in reference to the machine body temperature. In this setting, coolant temperature follows up the temperature of machine body.

Note : If the set temperature is too low, the spindle sleeve will be cooled excessively, causing large temperature differences between inner and outer races of the bearings. This will result in the seizure of the bearings.

Therefore, do not change the temperature setting carelessly.

If the spindle is used for high speed operation with the spindle cooling chiller set off, stop the spindle and allow the spindle to cool. After the spindle temperature has dropped sufficiently, turn on the spindle cooling chiller and restart the spindle.

4-3 Spindle Pneumatic Circuit

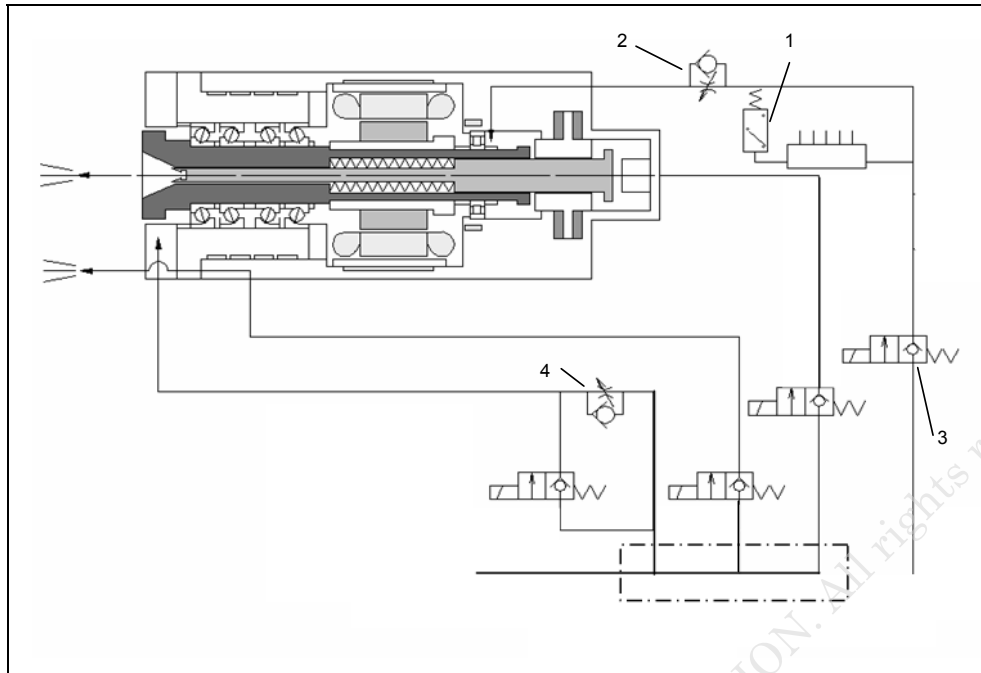


Fig. 4-3 Pneumatic circuit and pneumatic parts

No.	Part Name	Type	Maker
1	Pressure switch (air pressure of spindle lubrication)	IS10-01-L	SMC
2	Speed controller (for supplementary air supply for the rear bearing)	AS2201F-02-08	SMC
3	Solenoid valve (air for spindle lubrication)	VX2230-1D-B	SMC
4	Speed controller (air for labyrinth)	AS2201F-02-08	SMC

4-4 Spindle Hydraulic Circuit

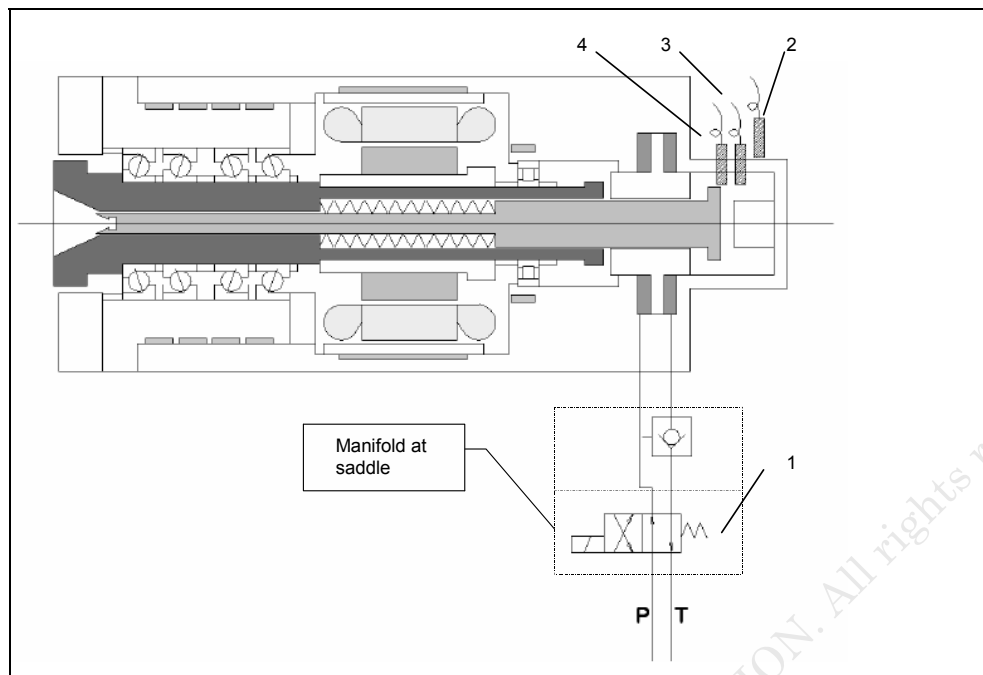


Fig. 4-4 Hydraulic circuit and hydraulic parts

No.	Part Name	Type	Maker
1	Solenoid valve (for tool unclamp)	SLD-G01-A3X-C1-30	FUJIKOSHI
2	Proximity sensor (for tool unclamp confirmation)	FL7M-3J6HD-CN03	Azbil
3	Proximity sensor (for clamp without tool in the spindle confirmation)	FL7M-2J6HD-CN03	Azbil
4	Proximity sensor (for clamp with tool in the spindle confirmation)	FL7M-2J6HD-CN03	Azbil

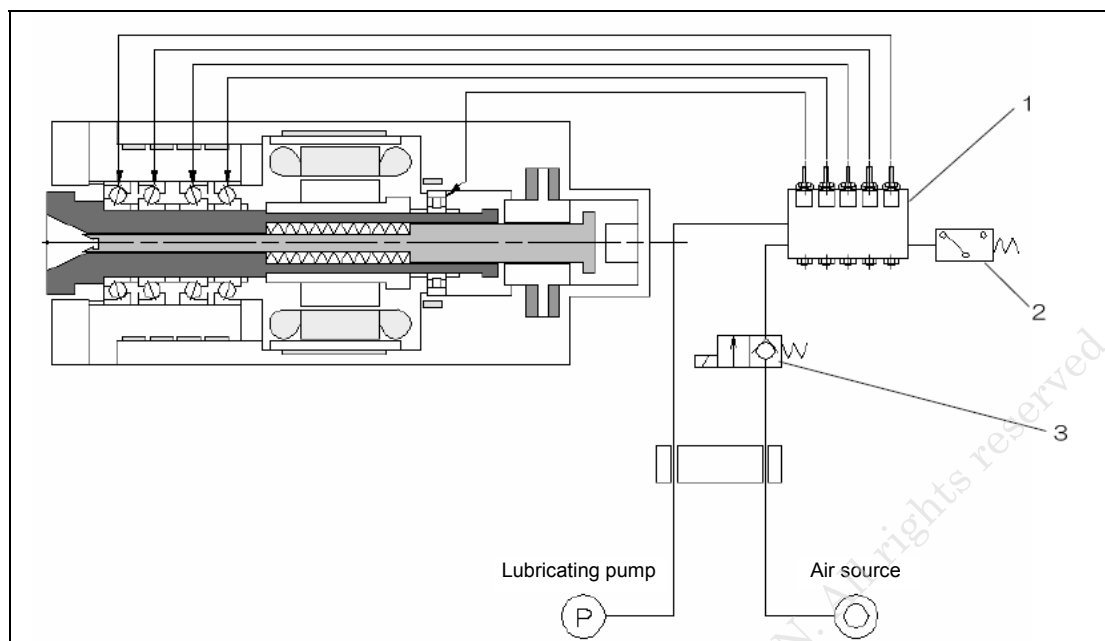


Fig. 4-5 Spindle lubrication circuit

No.	Part Name	Type	Maker
1	Mixing valve	MIX-1A4B	LUBE
2	Pressure switch	IS10-01-L	SMC
3	Solenoid valve	VX2230-02-1D-B	SMC

- | | |
|-------------------------------|---|
| Pump unit | Electric intermittent type gear pump |
| Motor | Power 100 VAC |
| | Output 19/18 W (50/60 Hz) |
| Pump Discharge | 90/110 cm ³ /min (5.49/6.71 in ³ /min) (50/60 Hz) |
| Discharge pressure | 1.5 MPa (220 psi) Relief valve setting pressure |
| Tank capacity | 1.8 L (0.06 ft ³) (effective) |
| Level switch | B-contact (OFF at lower limit of oil level) |
| Pressure switch | A-contact |
| Lubrication points | Spindle bearings |
| Operation interval | 8-minute cycle (OFF for 450 s and ON for 30 s) |
| Oil supply interval (approx.) | Approx. 230 days (in operations at 8 hours/day) |
| | Alarm output provided (by detecting the pressure) |

4-6 Arrangement of Electric Devices

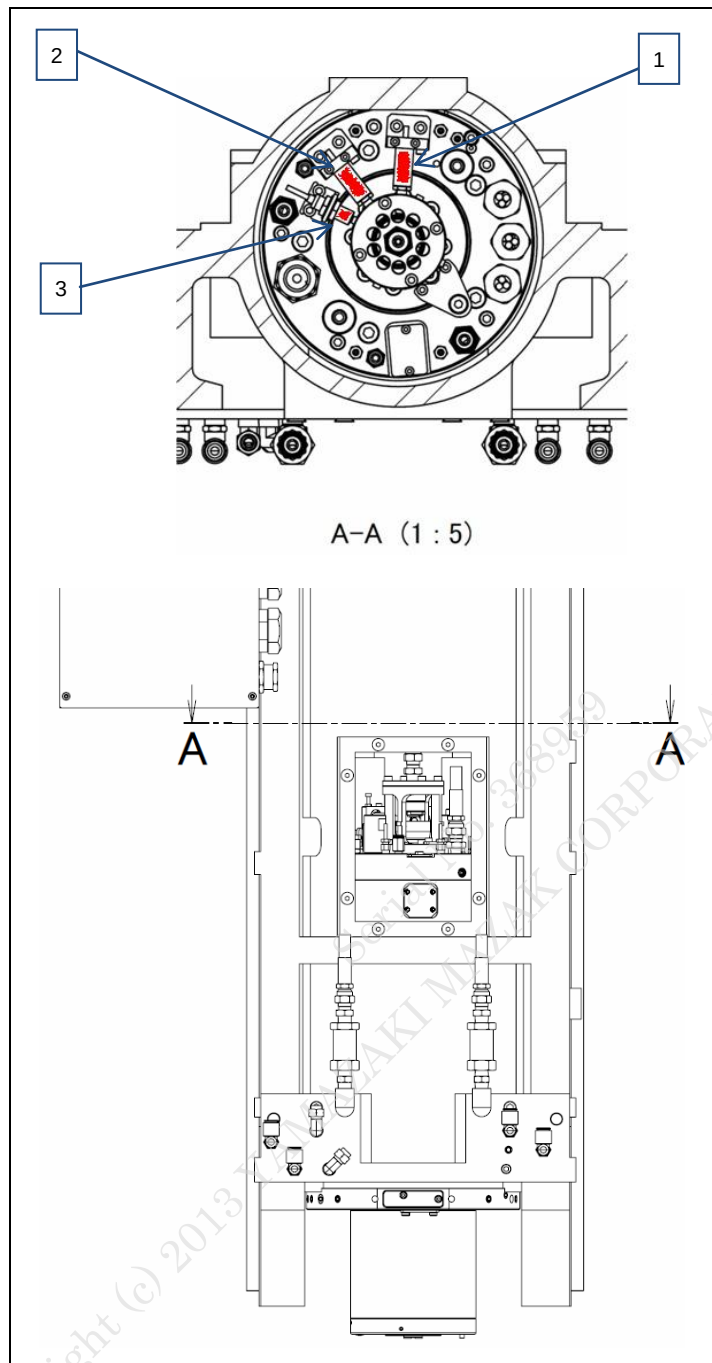


Fig. 4-6 Arrangement of electric devices

No.	Item	Use	Code
1	Proximity sensor	Tool unclamp confirmation	SQ10
2	Proximity sensor	Clamp without tool in the spindle confirmation	SQ12
3	Proximity sensor	Clamp with tool in the spindle confirmation	SQ11
4	Pressure switch	Low air pressure detection	SP5



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